

CFVBench: A Comprehensive Video Benchmark for Fine-grained Multimodal Retrieval-Augmented Generation

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Modalities

QA pairs

Existing Benchmarks



video



video-audio

CFVBench (ours)

\$7,832

\$2,031



audio-text-video

Q: What would happen if the person did not use the white substance dissolved in water in glass?

A: The dish would lack thickening agent.

Q: What instruments are in the video?

A: One woman is playing the violin and ... the large piano.

Q: For a \$10,000 debt at 13% interest, what are the exact total interest amounts paid under minimum payment and accelerated repayment scenarios?

A: Under minimum payments, the total interest paid is \$7,832; under accelerated repayment, the total interest paid is \$2,031.

Figure 1: Comparison of video-based MRAG benchmarks with CFVBench, where clues are embedded in tables or on-screen text of video frames, requiring fine-grained reasoning across comprehensive modalities and format (e.g., video, text, audio, chart, table, etc.).

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Abstract

Multimodal Retrieval-Augmented Generation (MRAG) enables Multimodal Large Language Models (MLLMs) to generate responses with external multimodal evidence. Recently, numerous video-based MRAG benchmarks have been proposed to evaluate model capabilities across retrieval and generation stages in MRAG. However, existing benchmarks remain limited in modality coverage and format diversity, often focusing on single- or limited-modality

tasks, or coarse-grained scene understanding. To address these gaps, we introduce **CFVBench**, a large-scale, manually verified benchmark constructed from 599 publicly available videos, yielding 5,360 open-ended QA pairs. CFVBench, spans high-density formats and domains such as chart-heavy reports, news broadcasts, and software tutorials, requiring models to retrieve and reason over long temporal video spans while maintaining fine-grained multimodal information. Using CFVBench, we systematically evaluate 7 retrieval methods and 14 widely-used MLLMs, revealing a critical bottleneck: current models (even GPT5 or Gemini) struggle to capture transient yet essential fine-grained multimodal details. To mitigate this, we propose Adaptive Visual Refinement (AVR), a plug-and-paly framework that adaptively increases frame sampling density and selectively invokes external tools when necessary. Experiments show that AVR consistently enhances fine-grained multimodal comprehension and improves performance across all evaluated MLLMs.

CCS Concepts

• **Information systems** → **Video search**.

Keywords

Multimodal Retrieval Augmented Generation, Video Understanding, Benchmark Evaluation, Multimodal Large Language Models

ACM Reference Format:

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1 Introduction

In recent years, the Retrieval-Augmented Generation (RAG) paradigm has become indispensable for enhancing the factual accuracy, knowledge grounding [27], and explainability [88] of Large Language Models (LLMs). This paradigm has been further extended to the Multimodal RAG (MRAG) framework, which leverages external multimodal knowledge (such as images, audio, and video) to address increasingly complex, real-world information-seeking tasks [43, 87]. Among these modalities, video-based MRAG [24, 38, 50, 60, 82] poses unique challenges due to its temporal dynamics, the tight coupling of visual and auditory streams, and the presence of dense, transient details [41].

To evaluate the increasingly sophisticated capabilities of video-based MRAG systems, numerous benchmarks [5, 7, 32, 78] have been proposed, typically focusing on tasks such as basic factual retrieval in visual questions, general video event understanding, or cross-modal reasoning. However, as illustrated in Fig. 1 and Table 1, existing benchmarks face two key limitations when confronted with real-world complexity. (1) **limited modality and format coverage**: current benchmarks often involve restricted modality pairings (e.g., text-video or audio-video), overlooking datasets that simultaneously encompass audio-text-video streams and densely embedded domain-specific formats, such as data charts, complex

Table 1: Comparison of video-based MRAG benchmarks. MR: Multimodal Fine-grained Reasoning; RQA: Retrieval-based Question-Answering; IM: Intra-Modal Multi-hop Questions; OE: Open-Ended Questions; CMF: Comprehensive Modalities and Format (e.g., video, text, audio, chart, table, etc.). Notation / indicates that benchmark is not presented in QA format.

Dataset	MR	RQA	IM	OE	CMF
Video Benchmarks					
ShareGPT4Video [5]	×	×	×	/	×
NExT-QA [74]	×	×	×	✓	×
Video-MME [16]	×	×	✓	×	×
LongVideoBench [73]	×	×	✓	×	×
MovieChat [59]	×	×	✓	✓	×
EgoSchema [40]	×	×	✓	×	×
MVBench [28]	×	×	×	✓	×
MMBench-Video [14]	×	×	×	✓	×
SOK-Bench [67]	×	×	×	×	×
Multimodal Benchmarks					
UnAV-100 [19]	×	×	/	/	×
VAST-27M [6]	×	×	✓	/	×
OmniBench [32]	✓	×	×	×	×
Cinepile [52]	×	×	×	×	×
Video-Bench [46]	×	×	×	✓	×
AVQA [75]	×	×	×	×	×
AVInstruct [78]	×	×	×	✓	×
Music-AVQA2.0 [35]	×	×	×	×	×
VGGSound [4]	×	×	/	/	×
AVHaystacks [10]	×	✓	✓	✓	×
AVHBench [61]	×	×	×	×	×
CFVBench (ours)	✓	✓	✓	✓	✓

tables, or dynamic on-screen text, which are critical for real-life video comprehension. (2) **insufficient fine-grained reasoning**: most benchmarks rely on coarse-grained analysis, such as scene classification or major entity recognition, and do not adequately evaluate a model's ability to perform precise, detailed multimodal reasoning. For example, extracting exact numerical values from rapidly changing charts or following sequential, understanding the complex operations and timing relationships in a tutorial, both of which are essential for grounded, factual multimodal question answering task.

To address these limitations, we introduce the Comprehensive Fine-grained Video-based retrieval-augmented generation Benchmark (*CFVBench*), a large-scale, manually verified, multimodal dataset specifically designed to evaluate video-based MRAG systems. *CFVBench*, is constructed from real-world videos on YouTube. It contains 599 videos and 5,360 open-ended question-answering (QA) pairs, spanning high-density content such as chart-heavy reports, news broadcasts, and software tutorials. The benchmark requires models to retrieve information across long temporal video spans and multiple modalities while maintaining a high level of fine-grained visual fidelity. Using *CFVBench*, we conduct a comprehensive evaluation of 7 popular retrieval methods and 14 publicly available Multimodal Large Language Models (MLLMs), revealing a critical bottleneck: *existing methods (even GPT5 or Gemini) struggle significantly with fine-grained multimodal comprehension*. Further

human evaluation shows that models frequently miss fine-grained visual cues and transient events, and exhibit internal issues such as hallucination, resulting in incomplete or inaccurate understanding.

Motivated by this critical finding, we propose the Adaptive Visual Refinement (AVR) framework. AVR is a plug-and-paly, two-stage framework that enables MLLMs to assess their current information state and execute a resource-efficient refinement strategy. In the first stage, adaptive frame interpolation mechanism employs a scoring system to evaluate information sufficiency and automatically increase frame sampling density only when an incomplete evidence chain is detected. In the second stage, on-demand tool invocation strategy intelligently activates specialized external tools, such as Optical Character Recognition (OCR) [66] and object detectors [8], when necessary to extract embedded visual text or entity relevant to the query. Experimental results show that AVR effectively enhances the fine-grained multimodal comprehension of existing MLLMs, enabling them to capture subtle visual details and achieve state-of-the-art (SOTA) performance on the proposed *CFVBench*. In summary, the main contributions of this work are:

- 1) We introduce *CFVBench*, a large-scale video-based MRAG benchmark (599 videos, 5,360 QA pairs) with high-density content (e.g., charts, tutorials), specifically designed to assess complex fine-grained multimodal comprehension across long temporal video spans.
- 2) We evaluate 7 retrieval methods and 14 MLLMs on *CFVBench*, uncovering a key bottleneck of existing MLLMs in fine-grained multimodal comprehension, likely caused by internal hallucinations and insufficient attention to transient details.
- 3) We propose AVR, a plug-and-paly framework that leverages adaptive frame interpolation and on-demand tool invocation to enhance fine-grained visual understanding and improve performance across all evaluated MLLMs. The dataset and code are available at <https://github.com/bluuueheart/CFVBENCH>.

2 Related Works

Multimodal RAG Benchmarks. Existing Multimodal Retrieval-Augmented Generation (MRAG) benchmarks have progressively expanded the integration of diverse modalities [43, 87], including text, images, audio, and video. Early knowledge-based visual question answering datasets, such as KB-VQA [69] and KVQA [58], rely on closed-domain knowledge, while OK-VQA [42] and A-OKVQA [57] incorporate external knowledge but mainly focus on single-step retrieval rather than complex reasoning. More recent benchmarks have broadened modality coverage and task complexity. MRAG-Bench [23] and MRAMG-Bench [79] combine text and images, Chart-MRAG Bench [76] integrates textual and chart data, Many-ModalQA considers text, images, and tables, WavCaps [44], and MusicCaps [1] emphasize audio-language understanding, KnowIT VQA [18] and SOK-Bench [67] focus on video reasoning, and InfoSeek [7] and Encyclopedic VQA [45] address knowledge-intensive, information-seeking tasks. Benchmarks with broader modality coverage, such as MultiModalQA [62] and AVHaystacks [10], still tend to simplify question formats through templates or multiple-choice settings (further comparison of the benchmarks are illustrated in Table 1.) Despite these advancements, most existing benchmarks remain limited in modality and format diversity and lack systematic

evaluation of fine-grained reasoning. To address this gap, we introduce *CFVBench*, a benchmark designed to evaluate fine-grained multimodal comprehension ability of different video-based MRAG systems.

Multimodal RAG Methods. Early MRAG methods typically converted multimodal data into unified textual representations for a text-based RAG pipeline [38, 82], causing substantial information loss, while later approaches preserved original multimodal data and introduced MLLMs for cross-modal retrieval and generation [60, 80]. Recent MRAG systems incorporate advanced modules such as multimodal search planners [36, 68], interleaved text-image output modules [29, 86], LLM-based cross-modal rerankers [34, 50], and cross-modal refiners [24, 72] to achieve end-to-end multimodal interaction. Despite these advances, evaluation still relies heavily on VQA [30] or VidQA datasets [56], limiting assessment of fine-grained reasoning. By conducting experiments on the proposed *CFVBench*, we find that existing video-based MRAG methods struggle with detailed multimodal comprehension, which motivates the Adaptive Visual Refinement (AVR) framework that evaluates retrieved visual information and selectively enhances frame sampling or invokes specialized tools to improve fine-grained comprehension.

3 CFVBench

3.1 Preliminary

Video-based MRAG is typically formulated as a two-stage process: (1) **Retrieval.** Given a textual question T_q , a retriever R selects a subset of videos (or video segments) $V = \{V_1, V_2, \dots, V_k\}$ that are most relevant to the question. The goal of this stage is to identify a minimal set of videos containing sufficient information to answer question T_q , thereby enabling efficient downstream processing. (2) **Generation.** A MLLM M then takes the question T_q and the retrieved video subset V as input, i.e., (T_q, V) , and synthesizes information across multiple modalities and temporal video segments to generate the final answer A . Formally, this can be expressed as

$$A = M(T_q, V) \quad (1)$$

To evaluate retrieval and generation while assessing fine-grained multimodal comprehension ability, we introduce *CFVBench*, a large-scale, manually verified benchmark of high-density videos.

3.2 Dataset Construction

As shown in Fig. 2, the construction of *CFVBench* follows a 4-stage pipeline to ensure diversity, difficulty, and quality:

Video Curation. We curated videos from YouTube spanning diverse domains such as finance, climate, and environment. Our focus was on content that inherently requires multimodal and fine-grained reasoning. The videos fall into 3 main categories: (i) *Structured-Data Videos*: Videos containing dense visual information such as charts and tables, which require accurate extraction of fine-grained details and numerical reasoning, aligned with transcripts. (ii) *Tutorials*: Instructional content (e.g., software or websites utilizing workflows) where UI operations are tightly coupled with audio scripts, demanding cross-modal reasoning. (iii) *News*: News reports that integrate live footage, scrolling captions, and occasional charts or texts, requiring models to capture objects, entities,

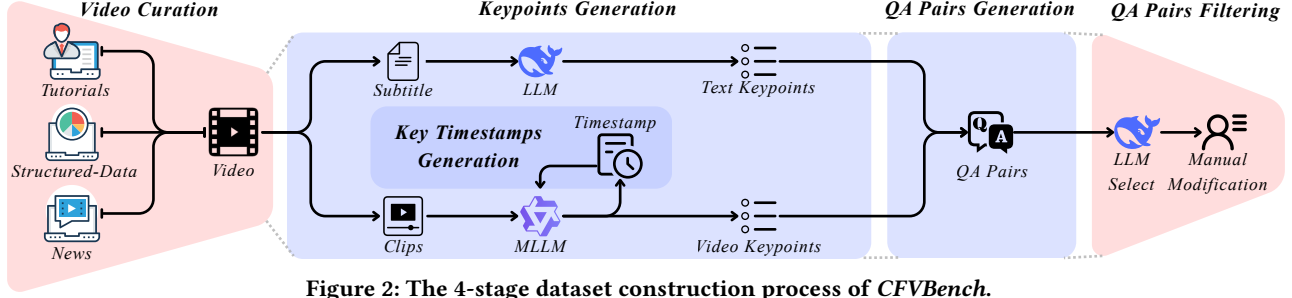


Figure 2: The 4-stage dataset construction process of CFVBench.

scenes, dynamics, and transient information. The specific websites corresponding to these categories are listed in Supplementary Material A. Each curated video is paired with its transcript and audio track, ensuring that *CFVBench* offers comprehensive multimodal coverage across video, text, and audio, as well as diverse format (such as charts and tables) in the video frames.

Keypoints Generation. To serve as the ground truth for QA construction, we extracted fine-grained, semantically independent facts (i.e., keypoints) from both the video frames and the textual subtitles. We employed MLLMs with domain-specific protocols to identify informative timestamps and extract visual, textual, or hybrid keypoints tailored to each video category (Structured-Data, News, and Tutorials), ensuring the necessity of cross-modal reasoning. The detailed methodology for this keypoint generation pipeline is provided in Supplementary Material C.

QA Pair Generation. Building on extracted keypoints, we constructed open-ended QA pairs with DeepSeek-R1. We generate (1) *single-hop* questions, answerable from a single keypoint (e.g., factual or numerical detail) from a single modality, and (2) *multi-hop* questions, requiring integrating two or more key points (e.g., sequential tutorial steps or combining visual and textual evidence) from multiple modalities. Questions primarily follow factual and procedural forms (“What,” “How”) while also including causal (“Why”), comparative (“How does ... compare”), and conditional (“What happens if ...”) patterns. Answers are grounded in keypoints without external inference, QA are independent and align with the general cognitive level of human video viewing, supporting reliable evaluation of retrieval and multi-step reasoning.

QA Pair Filtering. To ensure the quality and real-world applicability of the dataset, all generated QA pairs underwent a rigorous two-stage filtering process. Initially, we employed an automated pipeline using DeepSeek-R1 to filter pairs based on criteria such as keypoint necessity, topic relevance, and naturalness. Subsequently, to further enhance quality, all pairs were meticulously reviewed, verified, and modified by trained human annotators. This manual verification involved overlapping assignments and quality control checks to ensure inter-annotator consistency and dataset reliability. The complete filtering criteria and detailed human annotation protocol are available in Supplementary Material C.

3.3 Dataset Composition

After the dataset construction process, *CFVBench* consists of 3 core video categories: structured-data videos, tutorials and news. As shown in Fig. 3, these categories are further divided into topics including finance, climate, software operations, and commerce, covering key areas of real-world applications. Statistically, as reported in Table 8 of Supplementary Material B, *CFVBench* contains 5,360

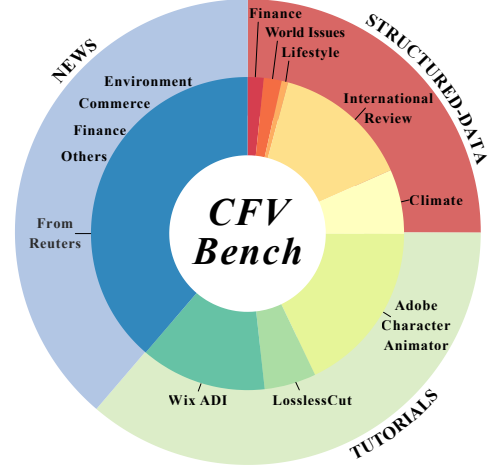


Figure 3: The distribution of videos in CFVBench.

question-answer pairs. To evaluate different levels of reasoning complexity, the test set includes 3,703 single-hop questions and 1,660 multi-hop questions. On average, multi-hop questions require integrating 2.72 keypoints across different modalities, placing higher demands on the model’s reasoning capabilities. The average video length is 232.3 seconds, and the average question length is 15.22 words, reflecting rich content and moderate question complexity.

4 Evaluation on CFVBench

MRAG frameworks are typically composed of two core stages: retrieval and generation. To thoroughly assess existing video-based MRAG methods’ performance, we separately evaluate both stages of MRAG on the proposed *CFVBench*. The following sections will first introduce the baselines, evaluation metrics, and implementation details relevant to both stages, followed by a comprehensive analysis of the experimental results.

4.1 Evaluation Settings

Baselines. The retrieval stage employs 3 widely-utilized multi-modal retrieval methods, including ImageBind [20], InternVideo [71], and LanguageBind [89] that align video and text representations within a shared embedding space. Building upon this, the nomic-embed-text [47] model is incorporated to specifically capture semantic similarities in the textual modality. After the retrieval stage, the top-*k* video clips identified by the best-performing retrieval method are passed to the generation stage.

During the generation stage, we combine the retrieval results and conduct a zero-shot evaluation on *CFVBench* with 14 MLLMs,

Table 2: Retrieval methods performance comparison under different scenarios.

Model	Overall				Multi-Point				Single-Point							
									Text-Single-Point				Video-Single-Point			
	R@1	R@3	R@5	R@10	R@1	R@3	R@5	R@10	R@1	R@3	R@5	R@10	R@1	R@3	R@5	R@10
Nomic-embed-text	51.01	64.53	69.42	76.11	41.74	56.20	60.84	68.19	58.33	71.01	75.74	82.13	49.26	63.12	68.62	75.05
Imagebind	37.16	54.03	61.21	70.63	25.42	38.97	45.18	54.91	42.12	61.23	68.82	79.97	42.98	59.95	67.62	74.67
Internvideo	30.05	45.12	52.32	63.21	20.00	29.27	35.36	43.73	34.16	52.03	59.70	71.97	35.32	52.59	60.34	71.88
Languagebind	46.18	61.88	68.13	76.17	32.16	43.37	48.67	57.28	53.93	71.84	78.48	86.31	49.72	67.07	73.81	81.48
Imagebind + nomic-embed-text	55.24	70.05	75.64	81.33	43.49	58.25	63.79	71.62	63.14	77.81	83.33	88.05	55.61	70.72	76.52	81.25
Internvideo + nomic-embed-text	54.09	67.55	72.83	78.76	42.83	56.74	62.65	69.06	61.69	74.83	79.35	85.15	54.37	67.85	73.74	79.24
Languagebind + nomic-embed-text	57.51	71.47	76.31	81.37	45.78	57.71	62.59	68.13	65.46	79.85	84.78	89.38	57.70	73.50	78.07	83.42

Table 3: Generation performance comparison of different MLLMs on CFVBench. The top 3 best scores for each metric are highlighted as first, second, and third. (red and blue to distinguish open-source and close-source models).

Model	Recall _t	Recall _v	Recall _{all}	Precision	F1	Rouge_L	St_cos	Likert	Fact_cov	Vis_use	Ling_prec
claude-opus-4	0.2031	0.2162	0.2079	0.1667	0.1850	0.0909	0.6545	3.2000	1.8909	2.3818	3.1273
gpt-5-chat	0.2079	0.1552	0.1887	0.1220	0.1482	0.1477	0.6705	3.5455	2.2386	2.3409	3.7045
gemini-2.5-flash	0.2843	0.3051	0.2919	0.1673	0.2127	0.1778	0.8000	3.6889	2.1000	2.5333	3.9444
Gemma-3-12b-it	0.0685	0.0601	0.0656	0.0837	0.0736	0.1287	0.6045	2.8205	1.4817	2.0272	3.0366
Qwen2.5-VL-7B-Instruct	0.1152	0.1220	0.1176	0.1067	0.1119	0.1376	0.6223	3.0996	1.6807	2.0953	3.7785
InternVL-3.5-8B	0.1805	0.2188	0.1936	0.1440	0.1652	0.1601	0.6814	3.4934	2.1734	2.3189	4.1022
llava-llama-3-8b	0.1583	0.2810	0.2018	0.0985	0.1324	0.2209	0.7933	3.5095	2.1308	2.2473	3.8312
MiniCPM-V-2_6	0.2407	0.3793	0.2892	0.1424	0.1908	0.1702	0.7766	3.6383	2.4286	2.3736	3.9560
InternVL3_5-14B	0.1858	0.2354	0.2034	0.1470	0.1707	0.2017	0.7272	3.7070	2.2843	2.3807	4.1494
Mistral-Small-3.2-24B-Instruct	0.1883	0.2041	0.1931	0.1069	0.1376	0.1639	0.6995	3.7104	2.2623	2.3880	4.0273
InternVL3_5-30B	0.2156	0.1818	0.2046	0.1193	0.1507	0.2000	0.7444	3.8296	2.4556	2.4000	4.1148
Intern-S1-mini	0.2941	0.2353	0.2745	0.1609	0.2029	0.2017	0.7647	3.8319	2.2797	2.3814	4.4153
Magistral-Small	0.2303	0.1512	0.2045	0.1057	0.1394	0.1883	0.7324	3.9351	2.5490	2.5621	4.1569
Gemma-3-27b	0.2655	0.1935	0.2400	0.1180	0.1582	0.2079	0.8020	3.9703	2.4851	2.5644	4.2277

including: Gemma-3-12b-it [63], Gemma-3-27b-it, Qwen2.5-VL-7B-Instruct [64], InternVL-3.5-8B [70], InternVL3_5-14B, InternVL3_5-30B-A3B, llava-llama-3-8b-v1_1 [11], Intern-S1-mini [3], MiniCPM-V-2_6 [77], Magistral-Small-2509 [51], Mistral-Small-3.2-24B-Instruct-2506, and 3 close-source models: gpt-5-chat-latest [48], gemini-2.5-flash-lite-preview-09-2025 [22], claude-opus-4-20250514 [2].

Evaluation Metrics. For retrieval evaluation, we adopt Recall@K [76] (R@K), which measures whether at least one relevant video is included among the top-k retrieved results. We report $k = 1, 3, 5, 10$ to capture performance across different cutoff thresholds. We evaluate two query types: (1) multi-point: queries require retrieving videos containing multiple keypoints across different modalities; (2) single-point: queries target videos with a single keypoint, further divided into text-single-point (speech/transcript-based, where keypoints are derived from audio transcripts) and video-single-point (visual content-based, where keypoints originate from visual information) to assess modality-specific retrieval capabilities.

In generation stage, following [76, 90], evaluation metrics are divided into two categories. (1) *Automatic evaluation*: To assess a model’s ability to leverage multimodal information in CFVBench, we adopt keypoint-based recall, precision, and F1-score [49]. Recall is computed for video ($Recall_v = C_v/N_v$) and textual keypoints ($Recall_t = C_t/N_t$), where C_v, C_t denote the number of correctly covered keypoints and N_v, N_t represent total number of ground truth keypoints, the overall recall is $Recall_{all} = (C_v + C_t)/(N_v + N_t)$. Precision measures the proportion of correctly covered keypoints among total claimed keypoints (M) in the response ($Precision = (C_v + C_t)/M$), and F1 is their harmonic mean. We also introduce

ROUGE-L [33] to capture lexical overlap. (2) *LLM-as-Judge evaluation*: We average the results from Qwen3-8B-Instruct [65] and GLM-4-9B [21] to holistically score model outputs. Each response is rated on three dimensions: Factual Coverage (Fact_cov), Visual Detail Usage (Vis_use), Linguistic Precision (Ling_prec), and assigned a final score on a 1–5 Likert scale [25]. We also compute semantic similarity (St_cos) between generated and reference answers using all-MiniLM-L6-v2 [53]. More implementation details, including hyperparameters, software versions, and baseline configurations, are provided in the Supplementary Material D.

4.2 Evaluation Results

Analysis of Retrieval Results. As shown in Table 2, we could find: combining textual and multimodal embeddings yields consistent improvements, with LanguageBind + nomic-embed-text achieving the best overall R@10 of 81.37%, underscoring the complementary strengths of text and video representations. In comparison, multi-modal embeddings alone generally outperform text-only retrieval on single-point queries, where the best video model (LanguageBind) achieves 76.17% R@10, slightly above the text baseline (76.11%). However, multi-point queries remain particularly challenging. An interesting observation is that multimodal embeddings underperform text-only embeddings under multi-point setting, suggesting that the video modality may introduce noise and hinder multi-hop reasoning. Moreover, across all multimodal methods, performance in multi-point setting lags behind single-point results, with even the best hybrid approach reaching only 71.62% R@10. This substantial

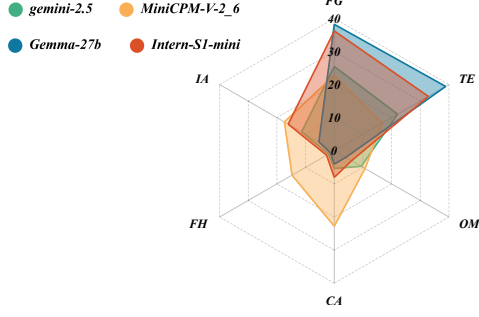


Figure 4: Human evaluation of typical MLLMs on CFVBench. FG: Fine-Grained Detail Omission, TE: Transient Event Neglect, OM: Object/Entity Misidentification, CA: Contextual Association Error, FH: Factual Hallucination, IA: and Incomplete Answer.

drop highlights that current models still struggle with fine-grained temporal integration and multi-hop reasoning.

Analysis of Generation Results. As shown in Table 3, the experimental results on CFVBench reveal several insights: (1) there are substantial performance differences among different MLLMs. For example, considering $Recall_t$, the highest-performing model achieves 0.2941 while the lowest is only 0.0685, indicating a large gap in the ability to extract key fine-grained information. (2) Although closed-source models that are not specifically designed for video tasks, such as gemini, do not show significant advantages over newer open-source video-oriented models, they demonstrate relatively stable performance across metrics, highlighting their effectiveness and reliability in video-based MRAG task. (3) The overall performance of all evaluated models remains modest across automatic and LLM-as-Judge evaluation, demonstrating the challenge posed by CFVBench and underscoring the necessity and importance of studying fine-grained video-based MRAG.

Manual Error Analysis. As shown in Fig. 4, we conducted a manual error analysis to further investigate model behavior. We randomly sampled 100 generated answers from Gemma-3-27B, MiniCPM-V-2_6, Gemma-27B and Intern-S1-mini, and asked 3 annotators to label each response according to 6 predefined error types. Majority voting was applied, and Fleiss’s kappa [15] of 0.81 indicated strong inter-annotator agreement. We observe: (1) all models frequently suffer from FG and TE errors, with Gemma-27B and Intern-S1-mini being most affected, highlighting the challenge of fine-grained information capture; (2) MiniCPM-V-2_6 produces fewer FG and TE errors but more CA, FH, and IA cases, suggesting that although it excels at detail perception, it is more vulnerable to hallucination and reasoning instability; (3) Gemini-2.5 demonstrates the most balanced performance, showing stronger robustness in fine-grained video-based MRAG task.

5 Methodology

To address the challenge of fine-grained comprehension, we propose Adaptive Visual Refinement (AVR) framework. First, the adaptive frame interpolation mechanism assesses evidence from sparse frame sampling and adaptively increases frame density for critical segments. Second, the visual refinement and generation strategy performs on-demand tool invocation, such as OCR [66] and object detection (DET) [8], and aggregates multimodal evidence to generate the final answer. The workflow of AVR is shown in Fig. 5.

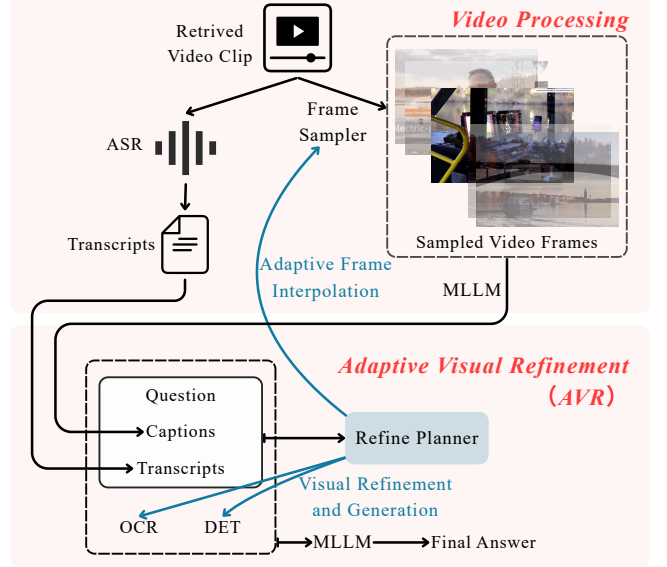


Figure 5: The workflow of the proposed AVR framework.

5.1 Adaptive Frame Interpolation

Adaptive frame interpolation evaluates the sufficiency of initial evidence and determines whether denser sampling is needed. For each retrieved video segment V_i , we first conduct video processing: (1) *Sparse Frame Sampling*: extract N_{sparse} frames (e.g., $N_{sparse} = 5$) to form a sparse frame set F_{sparse} . (2) *Audio Transcription*: generate subtitles T_{asr} using Automatic Speech Recognition (ASR) model Whisper [17]. (3) *Preliminary Summary*: obtain an initial summary C_{init} from F_{sparse} using the backbone MLLM.

We then feed the query T_q , subtitles T_{asr} , and summary C_{init} into the MLLM-based refinement planner with a refinement prompt, which explicitly instructs the model to assess whether the current evidence suffices to answer the query and to identify potentially missing information. The planner outputs an *Information Sufficiency Score* $S \in [0, 10]$, which integrates an overall *answerability score* measuring how well the query can be addressed given the current evidence and an *information density score* evaluating the richness and granularity of visual content in F_{sparse} and T_{asr} . A higher score S indicates sufficient evidence, while a lower score signals missing fine-grained details. Based on the information sufficiency score S , AVR determines the number of frames $N_{target} = f(S)$ to sample as:

$$f(S) = \begin{cases} N_{sparse} & \text{if } S > \theta \\ N_{min} + (N_{max} - N_{min}) \cdot \frac{\theta - S}{\theta} & \text{if } S \leq \theta \end{cases} \quad (2)$$

where θ is the threshold (set to 5.0) and $[N_{min}, N_{max}]$ is the refinement interval (set to $[20, 40]$; following Kandhare and Gisselbrecht [26] and Luo et al. [37], they both identify that the sampling rate achieves optimal results at around 1 fps). If $S > \theta$, the original sparse frames are retained; otherwise, denser sampling is applied proportionally to the severity of the information gap. Interpolated frames are filtered by visual similarity to reduce redundancy, and only diverse frames are retained for caption generation. Captions from these refined frames are merged with existing subtitles to construct a richer, fine-grained representation, which is subsequently fed into the generation stage of AVR.

Table 4: Generation performance comparison on CFVBench with the proposed AVR framework (marked with *).

Model	$Recall_t$	$Recall_v$	$Recall_{all}$	Precision	F1	Rouge_L	St_cos	Likert	Fact_cov	Vis_use	Ling_prec
claude-opus-4*	0.3125	0.3077	0.3103	0.2000	0.2432	0.1765	0.7059	3.6471	2.4706	2.8235	3.8824
gpt-5-chat*	0.2946	0.2727	0.2874	0.1644	0.2092	0.2211	0.7263	4.1158	2.8526	2.6632	4.0947
gemini-2.5-flash*	0.2210	0.4625	0.2950	0.1778	0.2219	0.1986	0.8582	3.7801	2.3901	2.5887	3.9929
Gemma-3-12b-it*	0.0621	0.0730	0.0660	0.1013	0.0799	0.1354	0.6247	3.0143	1.5879	2.1817	3.1508
Qwen2.5-VL-7B-Instruct*	0.1069	0.1449	0.1204	0.1049	0.1121	0.1513	0.6454	3.2943	1.6980	2.1510	3.9584
InternVL-3.5-8B*	0.1638	0.2941	0.2051	0.1457	0.1704	0.1667	0.7583	3.5542	2.2000	2.3958	4.1208
llava-llama-3-8b*	0.1523	0.3321	0.2160	0.1116	0.1472	0.2328	0.8017	3.5701	2.1490	2.3552	3.9380
MiniCPM-V-2_6*	0.2178	0.4561	0.3038	0.1846	0.2297	0.2111	0.8111	3.7111	2.4778	2.4333	4.0222
InternVL3_5-14B*	0.1692	0.2839	0.2099	0.1600	0.1816	0.2127	0.7485	3.8613	2.3691	2.4914	4.2580
Mistral-Small*	0.1857	0.2451	0.2035	0.1090	0.1420	0.1799	0.7143	3.7937	2.2910	2.4762	4.0952
InternVL3_5-30B*	0.2239	0.2941	0.2475	0.1217	0.1632	0.2712	0.8136	3.9831	2.5085	2.5339	4.1525
Intern-S1-mini*	0.2946	0.3667	0.3198	0.1993	0.2456	0.2626	0.8182	3.9495	2.3232	2.4343	4.4646
Magistral-Small*	0.2143	0.2958	0.2400	0.1176	0.1579	0.2598	0.7480	4.0000	2.6220	2.6220	4.2126
Gemma-3-27b*	0.2768	0.2581	0.2701	0.1247	0.1706	0.2500	0.8500	4.1000	2.5100	2.7800	4.2600

5.2 Visual Refinement and Generation

Guided by the MLLM-based refinement planner, we perform on-demand tool invocation and multimodal evidence aggregation to construct a rich feature representation for final answer generation. **On-demand Tool Invocation.** The refinement planner determines whether specialized tools are required. Specifically, OCR and DET tools are invoked only when explicitly required: OCR is used when the query depends on reading symbolic or structured visual data (e.g., numbers, UI elements, tables, charts), while DET is applied when answering requires recognizing entity categories, attributes, or spatial relations (e.g., object state, interactions, fine-grained classification).

If OCR is triggered, we apply EasyOCR [66] to the selected target frames F_{target} to extract symbolic or numeric information, such as textual overlays, tables, or chart values, and aggregate the results into T_{ocr} . If object detection (DET) is requested, we derive a task-specific detection vocabulary from the query T_q using a prompt-based keyword generator and run YOLO-World [9] to detect corresponding entities in F_{target} . The detection results (object class and spatial attributes) are converted into textual descriptions T_{det} .

Evidence Aggregation and Final Generation. For the selected target frames F_{target} , we employ an MLLM-based captioner to generate a detailed video summary C_{target} . All available evidence sources are then concatenated to form the enriched feature set $F_{rich} = [T_q, T_{asr}, C_{target}, T_{ocr}, T_{det}]$, where T_{ocr} and T_{det} are optional. Finally, the MLLM integrates F_{rich} to generate the final answer $A = M(T_q, F_{rich})$.

All MLLM-based components in AVR (including the refinement planner, captioner, and answer generator) employ the same backbone MLLM. The implementation details of AVR are shown in Supplemental Material D.

6 Experiment with AVR

Results with AVR. Table 4 demonstrates the effectiveness of AVR. Incorporating AVR leads to consistent metric improvements across all MLLMs, with particularly notable gains in $Recall_v$. For instance, MiniCPM-V-2_6 rises from 0.3793 to 0.4561, confirming AVR’s ability to enhance fine-grained multimodal comprehension through dynamic frame sampling and selective OCR/DET. These results suggest that AVR enables models to better capture critical visual details and reason over dense video content. A slight drop in $Recall_t$ for

Table 5: A performance comparison of various Video-RAG methods on CFVBench.

Baseline	Backbone	Likert	F1	St_cos
LLaMA-VID [31]	InternS1	1.7217	0.0723	0.3886
	Mistral	1.9904	0.0873	0.3896
	gemma27b	2.1232	0.099	0.4101
LLOVi [81]	InternS1	2.3358	0.0879	0.6076
	Mistral	2.9274	0.0829	0.7141
	gemma27b	3.0982	0.1055	0.7897
LongVA [84]	LongVA	2.7883	0.0957	0.6828
Flash-VStream [83]	Qwen	2.7978	0.0667	0.7199
	llava	3.1300	0.0929	0.7905
VideoAgent [13]	Mistral	3.1081	0.1189	0.6024
DrVideo [39]	InternS1	2.6466	0.0847	0.7108
	Mistral	3.1856	0.1137	0.8059
	gemma27b	3.5188	0.1105	0.841
DVD [85]	Mistral	3.5880	0.1736	0.7444
VideoRAG [54]	InternS1	3.1827	0.1143	0.6668
	Mistral	3.7654	0.1454	0.7517
	gemma27b	4.0500	0.1433	0.7957
AVR (ours)	InternS1	3.9495	0.2456	0.8182
	Mistral	3.7937	0.1420	0.7143
	gemma27b	4.1000	0.1706	0.8500

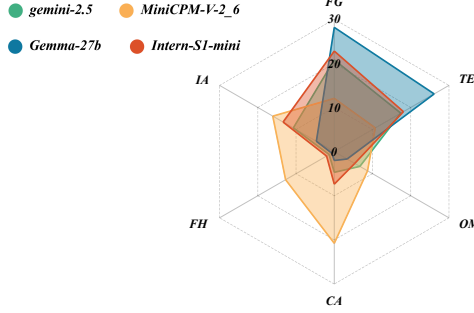
some models reflects the trade-off between richer visual coverage and potential retrieval noise.

We also conducted a comprehensive performance comparing AVR against a wide range of SOTA video understanding methods [13, 31, 39, 54, 81, 83–85]. To ensure a fair evaluation, all baselines were re-implemented under official protocols. The detailed results are presented in Table 5. The results demonstrate that AVR consistently outperforms all baselines across all evaluation metrics. Unlike existing methods that struggle to capture fine-grained details in complex video understanding tasks, AVR’s superior performance mainly stems from its adaptive frame interpolation and visual information refinement, which enable more precise and fine-grained multimodal comprehension.

Ablation Study. To assess the contribution of each AVR component, we perform ablation studies on two representative MLLMs:

Table 6: Ablation study on AVR, with green values showing the change in indicator scores relative to results w/o AVR.

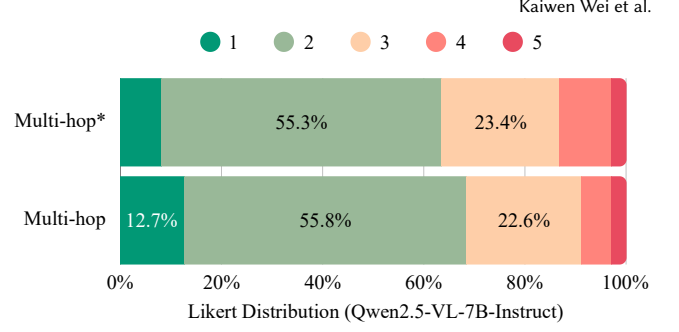
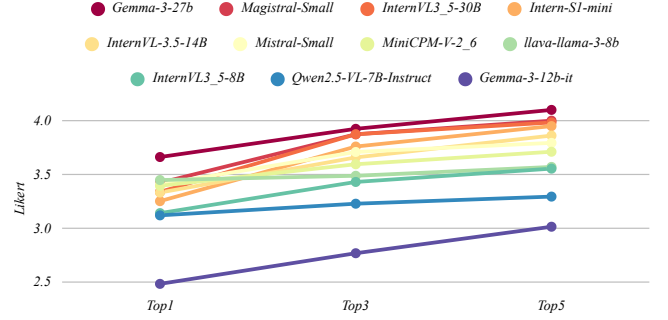
Model	$Recall_v$	Vis_use
Gemma-3-27b	0.1935	2.5644
Gemma-3-27b (+ frames)	0.2222 +14.83%	2.7059 +5.52%
Gemma-3-27b (+ frames & DET)	0.2407 +24.39%	2.7326 +6.56%
Gemma-3-27b (+ frames & OCR)	0.2545 +31.52%	2.7529 +7.35%
InternVL3_5-30B	0.1818	2.4000
InternVL3_5-30B (+ frames)	0.2593 +42.63%	2.4824 +3.43%
InternVL3_5-30B (+ frames & DET)	0.2778 +52.81%	2.5059 +4.41%
InternVL3_5-30B (+ frames & OCR)	0.2647 +45.60%	2.5085 +4.52%

**Figure 6: Human evaluation on CFVBench with AVR.**

Gemma-3-27b and InternVL3.5-30B. As shown in Table 6, all modules improve fine-grained multimodal comprehension. The adaptive frame interpolation strategy consistently boosts both $Recall_v$ and $VisUse$ by mitigating information loss through dynamic sampling. Further integrating OCR and DET tools yields additional gains, with OCR particularly effective for structured scenes involving textual or numeric content. Moreover, we conducted experiments to explore the performance on multi-hop questions and the impact of retrieval Top- k . See Supplementary Material 6 and 6 for the details.

Human Evaluation. Following the same protocol as Section 4.2, we further evaluated 100 responses from 4 representative MLLMs after integrating the AVR framework. Each response was independently annotated by 3 human raters, achieving a Fleiss's kappa [15] of 0.83, indicating strong agreement. Majority voting results are shown in Fig. 6, we find: (1) compared to methods without AVR in Fig. 4, both FG and TE errors of all MLLMs notably decreased, validating AVR's effectiveness in capturing fine-grained visual cues via adaptive frame selection and OCR/DET tools; (2) hallucination and contextual association errors persist, particularly for MiniCPM-V-2_6, suggesting that intrinsic reasoning instability remains even when visual details are accurately captured. A case study experiment was also conducted, please refer to Supplementary Material E for more details.

Performance on Multi-hop Questions. To further investigate model performance in complex scenarios that require integrating fine-grained visual cues for detailed reasoning, we utilize Qwen2.5-VL-7B-Instruct as backbone model and compare the 1–5 Likert [25] score distribution for multi-hop questions with (Multi-hop*) or without AVR (Multi-hop). Higher Likert scores indicate better balanced use of multimodal information and more complete coverage of key-points. As shown in Fig. 7, incorporating AVR leads to a noticeable increase in average Likert scores, demonstrating that AVR enables the model to better comprehend fine-grained content, which in turn supports more accurate multimodal multi-hop reasoning.

**Figure 7: Performance on multi-hop questions, where (*) indicates utilizing AVR.****Figure 8: Retrieval effect of different top- k settings.**

Impact of Retrieval Top- k . As shown in Fig. 8, we also conducted experiments comparing different retrieval top- k settings, using the Likert score as the evaluation metric. The Likert [25] score provides a holistic assessment of answer clarity, balanced use of multimodal information, and coverage of keypoints. We observe that as k increases from 1 to 5, the performance of nearly all models improves, indicating that retrieving more relevant segments is generally beneficial for the fine-grained video-based MRAG task. However, the magnitude of improvement varies across models, suggesting that different MLLMs have varying abilities to capture and utilize fine-grained multimodal information.

7 Conclusions

In this work, we introduce *CFVBench*, a large-scale, human-checked benchmark of 599 real-world videos and 5,360 open-ended QA pairs spanning tables, charts, and other formats. The benchmark covers audio-text-video modalities and is designed to evaluate fine-grained multimodal comprehension in existing video-based MRAG systems. Comprehensive evaluation of 7 retrieval methods and 14 MLLMs reveals that current approaches struggle with transient but critical multimodal details. To address this, we propose Adaptive Visual Refinement (AVR), a plug-and-paly framework that combines adaptive frame sampling with on-demand tool invocation, consistently enhancing fine-grained understanding and overall performance across all the evaluated MLLMs. *CFVBench* and AVR could provide a practical testbed and methodology for advancing video-based multimodal reasoning in real-world scenarios.

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Table 7: Data Source of CFVBench.

Topic	Website Link
Structured-Data Videos	Vox
Tutorials	Adobe Character Animator LosslessCut Wix ADI
News	Reuters

Table 8: CFVBench properties and question statistics.

Metric	Value
Average Video Length	232.3s
Average Question Length	15.22 words
Total Questions	5,360
Single-hop Questions	3,703
Multi-hop Questions	1,660
Average Keypoints per Multi-hop Question	2.72
What	2,581 (48.13%)
How	1,374 (25.62%)
Mixed	719 (13.41%)
Which	251 (4.68%)
Why	239 (4.46%)
Where	68 (1.27%)
When	59 (1.10%)
Who	30 (0.56%)
Other	42 (0.78%)

A CFVBench Data Source

The data source of different topics of CFVBench is in Table 7.

B Statistics of CFVBench

CFVBench is a large-scale, manually curated multimodal benchmark for evaluating fine-grained reasoning in video-based MRAG systems. As shown in Table 8, it contains 5,360 open-ended QA pairs from 599 real-world YouTube videos across diverse domains such as news, reports, and tutorials, with an average duration of 232.3 seconds. Questions are concise (15.22 words on average) and include 3,703 single-hop and 1,660 multi-hop items, the latter averaging 2.72 reasoning keypoints. “What” and “How” questions dominate (48.13% and 25.62%), while other types such as “Which,” “Why,” and “Where” contribute additional diversity.

C Dataset Construction Details

Keypoints Generation. (step 2) We used Qwen2.5-VL-72B-Instruct [64] to automatically identify informative timestamps under a category-specific protocol: For tutorial videos, the model segmented steps and annotated each operation concisely. For structured-data videos, it detects comprehensive data formats such as charts, tables, maps, and flowcharts, and outputs start and end times with formatted semantic tags in JSON. For news videos, salient frames were recognized and described at the image level (e.g., polar bears walking

on ice). All outputs were stored in a unified JSON schema with temporal boundaries and semantic annotations, providing reliable anchors for keypoint extraction.

Following [76, 90], keypoints are fine-grained, semantically independent facts from videos or subtitles, serving as ground truth for QA construction and evaluation. To handle diverse video domains, we designed tailored extraction protocols. For Structured-Data and News videos, we extracted (i) textual keypoints from subtitles via DeepSeek-R1 [12], and (ii) visual keypoints from frames aligned with timestamps using Qwen2.5-VL-72B-Instruct. For Tutorial videos, where narration and on-screen actions are tightly coupled, we generated hybrid keypoints from subtitles, frames, and timestamps, encoding explicit (action, object, outcome) triples (e.g., “the user clicks the ‘File’ button → the ‘Save As’ dialog box opens → a file path is displayed”). This framework ensures QA tasks require cross-modal reasoning rather than a single modality. To improve accuracy, we applied the LLMs three times on each sample and took the intersection of the results.

QA Pair Filtering. (step 4) To ensure the quality of generated QA pairs, we employed DeepSeek-R1 to automatically check each QA pair against 4 criteria: (i) all listed keypoints must be necessary to construct the complete answer for multi-hop questions, redundant or unused keypoints were removed, (ii) duplicates across QA pairs were eliminated, (iii) QA pairs unrelated to the topic were discarded, and (iv) overly mechanical or unnatural questions (e.g., over-detailing or forced associations) were filtered out.

To further enhance the quality of CFVBench and ensure questions are meaningful to humans and align with real-life application scenarios, we conduct human modification. We recruited 13 graduate students from local universities, all with academic backgrounds in computer science, journalism, or English, and IELTS scores above 7.0. Before annotation, they underwent training that introduced the task objectives and detailed guidelines (aligned with QA pair filtering criteria), demonstrated examples of both correct and incorrect annotations, emphasized data privacy and responsible usage, and clarified that they need to modify the QA pairs to make semantics more precise while keeping content unchanged. During annotation, each annotator was assigned a subset of QA pairs, with 30% overlapping between two annotators. Annotators were also instructed to adjust temporal segments when initial video timestamps were incorrectly split. A QA pair was finalized only when both annotators provided consistent annotations, ensuring dataset reliability. Additionally, for quality control, 2 supervising annotators randomly sampled 20% of every 100 QA pairs. If any issue was found, the process was repeated until all sampled pairs satisfied the standards.

D Evaluation Settings Details

Implementation Details. During the retrieval stage, we follow the procedure in [55], which leverages both audio transcriptions and video captions. Specifically, the audio of each video is transcribed using Faster-Distil-Whisper-Large-v3 [17], while visual frames sampled at 6-second intervals are captioned with MiniCPM-V-4-int4 [77], conditioned on the corresponding audio information. Based on these modalities, we generate multimodal embeddings from the raw videos and extract text embeddings from the generated descriptions. The top- k full videos are first retrieved using

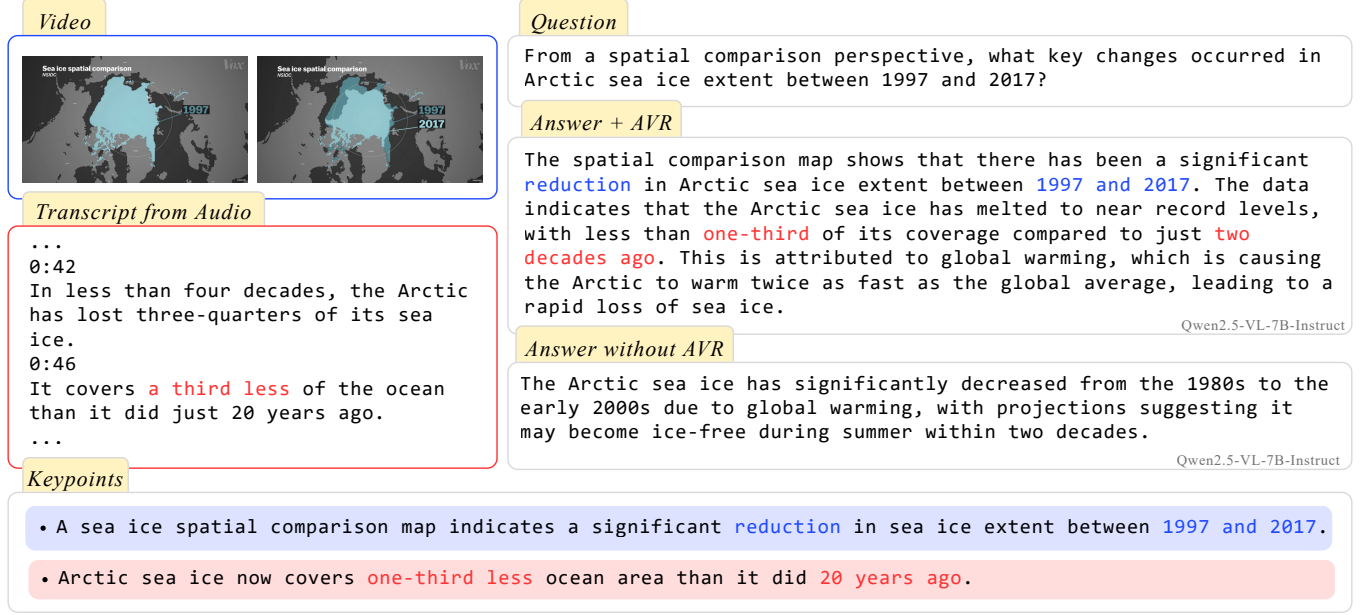


Figure 9: The case study experiment, where key clues are embedded in the on-screen text and audio transcript of the video.

cosine similarity, after which each is segmented into 30-second clips, and the results of best-performing retrieval method are then passed to the generation stage.

In the generation stage, we select the top 5 candidate clips. For each clip, its transcript is combined with 5 frames sampled at 6-second intervals, and the resulting multimodal input is fed into the MLLM in the retrieval order to generate the final answer. To ensure deterministic outputs, we set the temperature to 0.1 and top- p to 1. All MLLMs are evaluated using identical prompts on NVIDIA A100 GPUs with 80 GB of memory.

AVR Implementation Details. Our adaptive sampling strategy, detailed in Section 5, begins by extracting an initial, sparse set of frames, where $N_{sparse} = 5$. Based on the Information Sufficiency Score S generated by the refine planner, the system determines if denser sampling is needed. If S falls below the threshold $\theta = 5.0$, the target frame count N_{target} is proportionally calculated to fall within the refinement interval $[N_{min}, N_{max}]$, which is set to $[20, 40]$. We then sample an additional $N_{target} - N_{sparse}$ frames. These new frames are sampled uniformly from time segments within the video clip that are not in close proximity to the timestamps of the frames already selected. To ensure visual diversity and reduce redundancy, we apply a filtering process. We compute a perceptual hash (p-hash) for each frame. For every newly sampled frame, its hash is compared to the set of already retained frames. If the minimum Hamming distance to any retained frame is below a threshold (set to 5.0), the frame is considered redundant and discarded. The process is repeated, sampling from alternative time segments, until the target count of N_{target} visually distinct frames is reached. This final collection constitutes the target frame set, F_{target} .

As described in Section 5.2, the full set F_{target} is used by the MLLM-based captioner to generate the detailed summary C_{target} . However, to balance performance, processing speed, and model context limits, the on-demand tools operate on a *subset* of these

frames. Specifically, we uniformly sub-sample a set $F_{ocr} \subset F_{target}$ (where $|F_{ocr}| = 6$) for OCR and $F_{det} \subset F_{target}$ (where $|F_{det}| = 8$) for object detection.

Baseline Implementation Details. To establish comprehensive performance benchmark and validate the efficacy of AVR, we conducted a thorough evaluation of state-of-the-art video understanding models on *CFVBench*, encompassing both advanced VLMs and intelligent agent systems.

To ensure a fair comparison, all baseline methods re-implemented for this study adhere to the following specifications: (1) *Input Data*: We utilize the fundamental textual knowledge (e.g., visual captions and transcripts) derived from all videos, employing the same chunking protocol as our proposed method. (2) *Backbone Selection*: To ensure a fair comparison with AVR, we selected multiple backbones for each baseline where feasible. Notably, VideoAgent and DVD require models with tool-use (or tool-calling) capabilities; consequently, only Mistral was selected for these two methods. For LongVA and Flash-VStream, we utilized their officially provided checkpoints or models. (3) *Baseline Enhancements*: To accommodate the specific requirements of *CFVBench*, which performs RAG across the entire video collection, we augmented the vision-language models (LLaMA-VID, Flash-VStream, LongVA, and LLoVi) with the same retriever module used in AVR: LanguageBind + Nomic-Embed-Text.

E Case Study

To qualitatively illustrate AVR’s capabilities, we present a case study with Qwen2.5-VL-7B-Instruct (Figure 9), where key clues appear in on-screen text and the audio transcript. Without AVR, the baseline fails to link multimodal evidence and produces an incorrect answer, demonstrating hallucination. With AVR, the model successfully integrates multimodal cues to generate a complete, accurate, and coherent answer, showing that AVR effectively resolves incomplete evidence and enhances fine-grained visual comprehension.